```python
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

# Load the Iris dataset
iris = load_iris()
X = iris.data

# Applying K-Means Clustering
kmeans = KMeans(n_clusters=3, random_state=42)
kmeans.fit(X)

# Predicting the cluster for each data point
y_pred_clusters = kmeans.predict(X)

# Evaluating the model
inertia = kmeans.inertia_
silhouette = silhouette_score(X, y_pred_clusters)

print("Inertia:", inertia)
print("Silhouette:", silhouette)
```